

Source: Google Cloud Documentation, *MLOps: Continuous Delivery and Automation Pipelines in Machine Learning*

Section: MLOps Level 1: ML pipeline automation

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MLOps level 1: ML pipeline automation

The goal of level 1 is to perform continuous training of the model by automating the ML pipeline; this lets you achieve continuous delivery of model prediction service. To automate the process of using new data to retrain models in production, you need to introduce automated data and model validation steps to the pipeline, as well as pipeline triggers and metadata management.

The following figure is a schematic representation of an automated ML pipeline for CT.

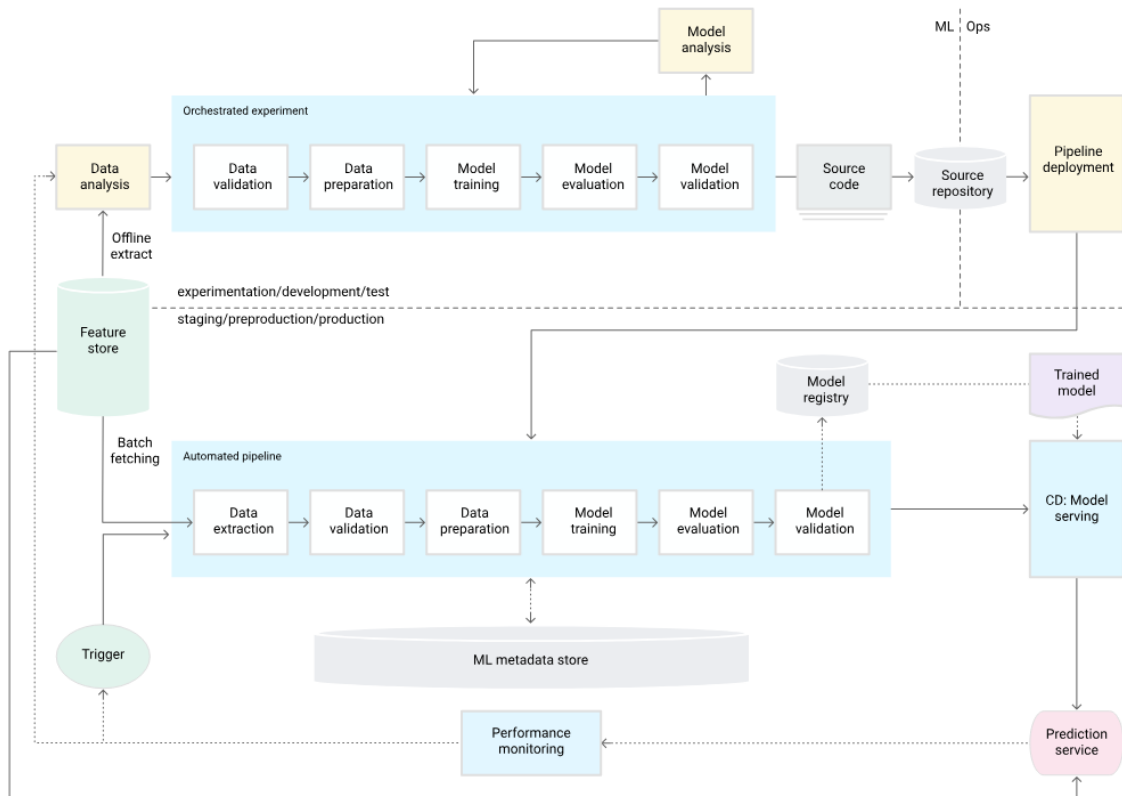


Figure 3. ML pipeline automation for CT.

Characteristics

The following list highlights the characteristics of the MLOps level 1 setup, as shown in Figure 3:

- **Rapid experiment:** The steps of the ML experiment are orchestrated. The transition between steps is automated, which leads to rapid iteration of experiments and better readiness to move the whole pipeline to production.
- **CT of the model in production:** The model is automatically trained in production using fresh data based on live pipeline triggers, which are discussed in the next section.
- **Experimental-operational symmetry:** The pipeline implementation that is used in the development or experiment environment is used in the preproduction and production environment, which is a key aspect of MLOps practice for unifying DevOps.
- **Modularized code for components and pipelines:** To construct ML pipelines, components need to be reusable, composable, and potentially shareable

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across ML pipelines. Therefore, while the EDA code can still live in notebooks, the source code for components must be modularized. In addition, components should ideally be containerized to do the following:

- Decouple the execution environment from the custom code runtime.
 - Make code reproducible between development and production environments.
 - Isolate each component in the pipeline. Components can have their own version of the runtime environment, and have different languages and libraries.
- Continuous delivery of models: An ML pipeline in production continuously delivers prediction services to new models that are trained on new data. The model deployment step, which serves the trained and validated model as a prediction service for online predictions, is automated.
 - Pipeline deployment: In level 0, you deploy a trained model as a prediction service to production. For level 1, you deploy a whole training pipeline, which automatically and recurrently runs to serve the trained model as the prediction service.